

Project Report: Design and Implementation of an Analog Artificial Neuron Model using Operational Amplifier

1. Abstract

This project presents the hardware implementation of an **Analog Artificial Neuron Model** using the **TL074 Quad Operational Amplifier**. In modern computational intelligence, Artificial Neural Networks (ANNs) are usually simulated via software. However, this project explores the underlying physical principles by constructing a physical neuron using analog electronics. The circuit successfully demonstrates the three fundamental properties of a biological and computational neuron: **Input Weighted Summation, Biasing (Offset Control), and Threshold Activation (Decision Making)**. The working model was verified using a function generator, dual DC power supply, and an oscilloscope.

2. Introduction & Theoretical Background

An artificial neuron is a mathematical function conceived as a model of biological neurons. The basic mathematical expression for a single-layer neuron (Perceptron) is:

$$y = f\left(\sum_{i=1}^n W_i X_i + b\right)$$

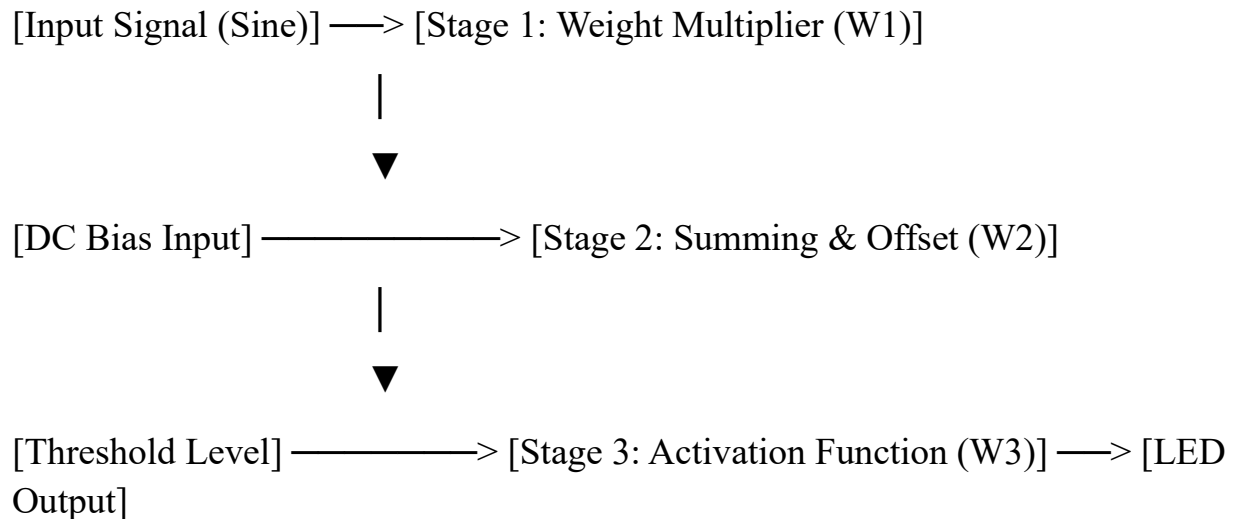
Where:

- X_i = Input Signals (represented here by an AC Sine wave)
- W_i = Weights (represented by Potentiometer 1)
- b = Bias/Offset (represented by Potentiometer 2)
- $f(\bullet)$ = Activation Function (represented by the Op-Amp Comparator and Potentiometer 3)
- y = Final Output (0 or 1, represented by the LED state)

In this analog model, we convert these mathematical operations into voltage levels using Operational Amplifiers.

3. System Architecture & Block Diagram

The implemented neuron is divided into three consecutive functional stages:



4. Detailed Circuit Description

The complete system is built around a single **TL074 Quad Op-Amp IC**, which contains four independent operational amplifiers. We utilized three stages of this IC:

Stage 1: Input Weight Multiplier (Op-Amp 1)

- **Configuration:** Inverting Amplifier.
- **Operation:** The input 1 kHz sine wave from the function generator is fed into the inverting terminal (Pin 2). Potentiometer 1 (W_1) is placed in the feedback loop or input path to change the overall gain ($A_v = -R_f/R_{in}$). By rotating this pot, the amplitude (size) of the input signal is scaled, mimicking the strengthening or weakening of a neural synapse.

Stage 2: Summing Junction & Bias Insertion (Op-Amp 2)

- **Configuration:** Inverting Summing Amplifier.
- **Operation:** This stage takes the weighted AC signal from Stage 1 and adds a variable DC voltage controlled by Potentiometer 2 (W_2). Mathematically, it performs the addition of $\sum W_i X_i + b$. On the oscilloscope, this causes the

entire sine wave to shift vertically **Up and Down**, altering the baseline energy level of the neuron.

Stage 3: Activation Function / Comparator (Op-Amp 3)

- **Configuration:** Open-Loop Voltage Comparator (Step Activation Function).
- **Operation:** The shifted sine wave is fed into one input of this op-amp, while Potentiometer 3 (W_3) provides a fixed reference/threshold voltage to the other input.
 - When the instantaneous AC signal voltage rises **above** the threshold set by W_3 , the op-amp saturates to its positive rail ($+V_{cc} \approx +9V$), delivering current to turn the **Red LED ON** (Logic 1).
 - When the signal falls **below** the threshold, the op-amp drops to its negative rail ($-V_{ee} \approx -9V$), reverse-biasing the diode and turning the **Red LED OFF** (Logic 0).

5. Bill of Materials (BOM)

SL.	Component Description	Specifications / Part No.	Qty	Unit Price (BDT)	Total Price (BDT)
1	Quad Op-Amp IC	TL074 / LM324	1	25.00	50
2	Potentiometer (Rotary)	10 k Ω Linear	3	15.00	45.00
3	Carbon Film Resistor	10 k Ω , 0.25 W	4	1.00	4.00
4	Carbon Film Resistor	1 k Ω , 0.25 W	2	1.00	2.00
5	Light Emitting Diode (LED)	5 mm Red LED	1	2.00	2.00
6	Breadboard	Standard 840-Point	1	Existing	Existing

SL.	Component Description	Specifications / Part No.	Qty	Unit Price (BDT)	Total Price (BDT)
7	Connecting Wires	Single-core Jumper Wires	1 Set	30.00	
—	Total Budget	—	—	—	133 BDT

6. Experimental Observations & Real-world Analysis

During the laboratory testing, the hardware exhibited the following behavioral responses:

1. **Vertical Wave Shifting:** Adjusting Potentiometer 2 (W_2) successfully introduced DC offset, shifting the sine wave dynamically along the Y-axis of the oscilloscope.
2. **State Transition:** Tuning Potentiometer 3 (W_3) altered the threshold level, causing the output LED to switch between discrete **ON** and **OFF** states, confirming the execution of a hard-limiting activation function.
3. **Hardware Constraints Encountered:** Real-world parasitics such as internal breadboard row capacitance and oscilloscope probe impedance affected high-speed edge transitions. These non-idealities caused slight smoothing of the output square wave at Pin 7. However, the binary switching logic was fully validated by the physical response of the load (LED).

7. Conclusion

The implementation of the analog artificial neuron successfully bridges the gap between software computational intelligence and hardware circuit design. The project demonstrates that complex mathematical neural nodes can be efficiently replicated using simple, cost-effective analog components like operational amplifiers and linear pots. The circuit performs robustly under standard laboratory conditions and satisfies all theoretical boundary conditions.